

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

ANSWERS:

1. Missing OSI Layers

Question:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Concept Map:

Application



Presentation



Session



Transport



Network



Data Link



Physical

Analysis:

- Missing layers are Session and Data Link.
- Session layer establishes, manages and terminates sessions.
- It provides synchronization and recovery.
- Data Link layer performs framing, MAC addressing and error detection.
- Together they improve reliable communication.

2. Data Units

Concept Map:

Application



Presentation



Session



Transport → Segment



Network → Packet



Data Link → Frame

↓
Physical → Bits

Analysis:

Data is encapsulated as Segment, Packet, Frame and Bits while moving down the OSI model. At the receiver it is decapsulated back to application data.

3. MAC vs IP

Concept Map:

Data Link → MAC Address → Local Delivery

Network → IP Address → Global Delivery

Analysis:

MAC addresses deliver frames within the local network. IP addresses route packets across different networks. Together they provide end-to-end communication.

4. Error Control

Concept Map:

Error Control

- └─▶ Data Link Layer → Error Detection (CRC)
- └─▶ Transport Layer → Error Correction (ACK, Retransmission)

Analysis:

The Data Link layer detects corrupted frames while the Transport layer retransmits lost data and guarantees reliable delivery.

5. Web Browsing

Concept Map:

Application (HTTP)

↓

Presentation (Encryption/Compression)

↓

Session (Session Management)

↓

Transport (TCP)

↓

Network (IP)

↓

Data Link (Ethernet/Wi-Fi)

↓

Physical (Signals)

Analysis:

Failure in any layer affects upper layers. For example, if the Network layer fails, TCP cannot deliver data and the browser cannot load the webpage.

6. Troubleshooting

Concept Map:

Physical ✓



Data Link ✓



Network ✗



Transport ✗



Application ✗

Analysis:

Physical and Data Link are functioning. Failure starts at the Network layer due to routing/IP issues. Since upper layers depend on lower layers, Transport and Application also fail. This concept map helps identify the faulty layer quickly.